

Programming 2025/2026

Laboratory

Session 2 (April 17)

While Loop

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While Loop

- The condition is a Boolean expression which is evaluated to **True** or **False**
- If the condition is **True**, the program will **execute** the expressions inside the **while** statement (indentation!!!)
- After that, the condition is evaluated **again**
- If the condition is **True**, the program will **again** execute the expressions inside the **while**
- And so on...until the condition is **False**
- If the condition is **False**, the program does not anymore execute the expressions inside the while and the execution passes to the expressions outside the **while**

```
while( condition ):
    expressions inside the while
    ...
    expressions inside the while
expressions outside the while
```

While Loop

- The number of iterations is determined by the variables involved in the while loop
 - Much more flexible than a `for` loop
 - Needs initialisation as in the `for` loop
- **WARNING:** The variables checked in the condition must be modified inside the while statement
 - What happens otherwise?

```
a = 3
i = 0
while( i <= 0 ):
    print(a)
```

Exercise (5.1)

- Write a program that prints all the vowels contained in a given string, by using a while loop

Exercise (5.2)

- Write two versions of a program that prints all the divisors of a given number, one using a for loop and the other using a while loop

Exercise (5.3)

- Write a program that prints the number of punctuation characters in a given string, by using a while loop

Exercise (5.4)

- Write a program that generates a random number between 1 and 100 and asks the user to guess it; prints “too big” or “too small” for every wrong guess and continues to ask until the correct number is found. Once found, the program reports the number of guesses.
- HINT: Use a function from the random module to generate a random number between two given numbers